

Fake News Detection

Project Overview

Fake News Detection is a Machine Learning project developed using Python to classify news articles as **Fake** or **True**. The project uses **Logistic Regression** to train a model and predict the authenticity of news.

Technologies Used

- Python
- Jupyter Notebook
- Pandas
- NumPy
- Scikit-learn
- Logistic Regression

Dataset

The dataset contains two types of news: **Fake.csv** – fake news articles, and **True.csv** – true news articles. Due to the large dataset size, the original CSV files are not included in this GitHub repository.

Project Workflow

- 1 Load the dataset
- 2 Clean and preprocess the news data
- 3 Combine the Fake and True news datasets
- 4 Split the data into training and testing sets
- 5 Convert text into numerical features using TF-IDF
- 6 Train the Logistic Regression model
- 7 Evaluate the model
- 8 Predict whether a news article is Fake or True

Model Used

Logistic Regression

The trained model is used to classify news articles into two categories: Fake and True.

Project Files

- **Fake_News_Detection.ipynb** – Main Jupyter Notebook containing the complete project
- **README.md** – Project documentation

- **requirements.txt** – Required Python libraries

Conclusion

This project demonstrates how Machine Learning and Natural Language Processing techniques can be used to identify fake news and classify news articles based on their content.